

KLH UNIVERSITY

Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester(2025-26)
Faculty Name	Ms Md. Razia
Student Name	Adunuri Bhavana sri
Roll Number	2520030232
Branch	CSE
Course Outcome	4

Case Study Topic: HealthNet – Intelligent Hospital Resource Management System using Dijkstra's Algorithm

Work Done Summary:

- Implemented Dijkstra's Shortest Path algorithm in Java to optimize the emergency transportation latency and clinical logistics routing paths between regional medical centers, critical supply reserves, and receiving hospitals.
- Modeled medical centers, ambulance hubs, and central pharmacy deposits as explicit nodes within a weighted undirected network topology graph.
- Utilized dynamic edge weights representing traffic density, physical distance, and emergency travel time overhead parameters.
- Supported instant shortest-path computations to safely direct critical life-support resources toward emergency crisis points without routing starvation.
- Compared single-source shortest path execution correctness across distinct graph weights to ensure system performance guarantees.

Implementation Details:

- **1. Configured NetworkGraph class** utilizing an optimized adjacency list framework to safely sustain complex multi-weight node configurations.
- **2. addRoute()** binds clinical nodes together with explicit cost attributes mapping transit latency values.

- **3. computeDijkstra(startNode)** isolates the shortest route configuration mapping from an origin clinical point using a Min-Priority Queue structure.
- **4. Distance update array** manages running shortest-path values dynamically while updating paths to minimize traversal overhead loops.
- **5. Main routine simulation** seeds regional emergency hubs, registers latency paths, executes node route requests, and outputs optimal transit path results.

Result: Screenshots/Output

```
PS D:\JavaPrograms> javac HealthNetDijkstra.java
PS D:\JavaPrograms> java HealthNetDijkstra

HealthNet - Emergency Transit & Resource Routing Engine (CO4)
Initializing Clinical Infrastructure Adjacency Graph...

Registered Weighted Transit Pathways:
Station 0 (Ambulance Hub) <-> Station 1 (Trauma Center)      [Travel Time = 4 mins]
Station 0 (Ambulance Hub) <-> Station 2 (Blood Repository) [Travel Time = 8 mins]
Station 1 (Trauma Center)  <-> Station 3 (Central ICU)       [Travel Time = 3 mins]
Station 2 (Blood Repository) <-> Station 3 (Central ICU)     [Travel Time = 2 mins]

Calculating Single-Source Shortest Path from Station 0 via Dijkstra's Algorithm:
Minimum Transit Time to Station 1 (Trauma Center): 4 minutes via Path [0 -> 1]
Minimum Transit Time to Station 3 (Central ICU): 7 minutes via Path [0 -> 1 -> 3]

System Integrity Log: All pathways checked. Critical transit configuration optimized.
```

Time Complexity & Empirical Analysis:

- Dijkstra's Execution Timeline: $O((V + E) \log V)$ utilizing a binary heap-based Priority Queue.
- Path Matrix Instantiation Space: $O(V + E)$ to preserve the structural boundaries of adjacent node link pairs.
- Dijkstra's structure guarantees mathematically sound transit pathway selection over naive unweighted traversal schemas for critical networks.

GitHub Repository Link:

<https://github.com/bhavana-sri25/Sketch-DSA-CO4>

Student Signature

Faculty Signature